

CART SURE

E commerce
application

SOFTWARE TEST PLAN DOCUMENT

Prepared by : Mohammed
Qadeer

Version : 1.0

Date :

Testing type : manual and
automation testing

Table of content

- **1. Introduction**
- **2. Business Scenario**
- **3. Scope of Testing**
- **4. Objectives**
- **5. User Journey Testing**
- **6. Test Strategy**
- **7. Testing Types**
- **8. Functionality Testing Checklist**
- **9. Test Environment**
- **10. Entry & Exit Criteria**
- **11. Resource Planning**
- **12. Testing Schedule**
- **13. Test Deliverables**

- **14. Assumptions & Dependencies**

- **15. Defect Management Process**

- **16. Tools Used**

- **17. Approval**

1 Introduction.

The purpose of this document is to prescribe the scope, framework, and resources needed to execute testing for [cart sure e commerce application]. This

plan defines the entry
and exit quality
benchmarks

required to achieve project
sign-off and ensure a defect-
free deployment.

2 BUSINESS SCENARIO

CartSure is a real-time eCommerce platform where users can browse products, search items, add products to cart, proceed to checkout, complete online payments, and receive order confirmations.

The QA team is responsible for validating all business-critical functionalities before deployment.

3 Scope of testing

In Scope:

- Login & Signup Functionality
- Product Listing & Search
- Cart & Checkout Functionality
- Payment Gateway Integration
- Order History & Confirmation
- UI/UX Validation
- Responsive Testing

Out scope :

- Third-party payment gateway internal logic
- Logistics backend systems
- Infrastructure & server-side performance tuning

4 Objectives

The primary objective of e-commerce testing is to ensure a secure, seamless, and uninterrupted shopping experience from product discovery to checkout. It validates that the platform meets business requirements, prevents revenue loss, and protects sensitive customer data.

Validate Functionality

Verify Checkout and Payments

Ensure Data Security

Evaluate Performance and Scalability

5. User Journey

Testing

The following user journey will be validated during testing:

1. User opens website
2. User searches for Shoes
3. User views product details
4. User adds product to cart
5. User proceeds to checkout
6. User enters shipping details
7. User completes payment
8. User receives order confirmation

6. Test Strategy

Testing activities will follow the Software Testing Life Cycle (STLC), including requirement analysis,

test planning, test case development, environment setup, test execution, defect reporting, regression testing, and final sign-off.

7. Testing Types

1. Core Functional Testing

- **Functional Testing:** Verifies that links, search filters, shopping carts, and checkout flows operate according to specifications.
- **Integration Testing:** Checks if data transfers seamlessly between the website, inventories, CRM systems, and third-party fulfillment logistics.
- **Payment Gateway Testing:** Validates transactions, handling fees, refunds, and declined card responses using tools like Postman.
- **API Testing:** Confirms the integrity of third-party plugins handling taxes, real-time shipping quotes, and order histories.
- **Regression Testing:** Ensures new updates, promotions, or bug fixes do not accidentally break existing core functions.

2. User Experience & Design Audits

- **Usability Testing:** Gauges how easily a first-time visitor can navigate from your homepage to order placement.
- **Compatibility Testing:** Assures the layout scales perfectly across mobile phones, tablets, and varying operating systems.
- **Cross-Browser Testing:** Verifies consistent look and function on Chrome, Safari, Edge, and Firefox via platforms like BrowserStack.

- **A/B Testing:** Compares two design iterations of a high-impact element (e.g., CTA button color) to optimize conversion rates.
- **Accessibility Testing:** Checks compliance with digital accessibility standards (like WCAG) to ensure everyone can navigate your site

3. Stability & Infrastructure Security

- **Security Testing:** Screens for database vulnerabilities like SQL injections, cross-site scripting (XSS), and data leakage using tools like OWASP ZAP.
- **Load & Stress Testing:** Simulates immense concurrent user traffic via Apache JMeter to prevent site crashes during massive flash sales or holiday shopping peaks.

4. Market Customization

- **Localization Testing:** Checks if multi-regional sites correctly display local languages, precise currency conversions, local tax configurations, and regionally compliant shipping parameters.

8. Functionality

Testing Checklist

Home Page: Navigation menu, featured products, banners.

Product Page: Product title, images, filters, sorting.

Cart Module: Add/remove/update cart items.

Checkout: Address form validation and order summary.

Payment: Payment success and failure scenarios.

Information Pages: Privacy Policy, Returns Policy, About Page.

9. Test Environment

Operating Systems: Windows 10/11

Browsers: Chrome, Firefox, Edge

Devices: Desktop, Tablet, Mobile

Automation Tools: Selenium, TestNG

Defect Tracking Tools: Jira/Bugasura

10. Entry & Exit

Criteria

Entry Criteria:

- Requirements finalized
- Build deployed successfully
- Test environment ready
- Test data prepared

Exit Criteria:

- All critical test cases executed
- No high severity defects open
- Regression testing completed
- Final report approved

11. Resource Planning

1. Personnel & Roles

Determine your core testing team based on project scale and timelines:

- **QA Lead/Test Manager**: Defines the testing scope, prioritizes features, and manages the overall test plan.
- **Functional Testers**: Execute user journey testing (search, add-to-cart, checkout, guest login).
- **Automation Engineers**: Build and maintain scalable, automated test scripts for routine regression.
- **Performance Testers**: Simulate high-traffic scenarios (e.g., flash sales or holiday shopping) to prevent server crashes.
- **Security Specialists**: Conduct penetration testing and compliance checks to protect user data.

12. Testing Schedule

- . Requirement Analysis – 2 Days
- . Test Planning – 2 Days
- . Test Case Design – 4 Days
- . Environment Setup – 2 Days
- . Test Execution – 7 Days
- . Regression Testing – 3 Days
- . Final Test Report – 1 Day

13. Test Deliverables

- Test Plan Document
- Test Strategy Document
- Test Scenarios & Test Cases
- Requirement Traceability Matrix (RTM)
- Defect Reports
- Automation Scripts
- Final Test Summary Report

14. Assumptions & Dependencies

Assumptions: Stable environment, finalized requirements, and available test data.

Dependencies: Application build availability, developer support, and payment gateway sandbox access.

15. Defect Management Process

All identified defects will be logged into Jira/Bugasura with appropriate severity and priority levels.

After fixes are provided by developers, QA team will perform retesting and regression testing.

16. Tools Used

- Selenium – Automation Testing
- TestNG – Test Framework
- Jenkins – CI/CD Integration
- Apache JMeter – Performance Testing
- Jira/Bugasura – Defect Tracking
- GitHub – Version Control

17. Approval

This document requires approval from QA Lead, Project Manager, and Client Stakeholders before testing execution begins.